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AmpC carrying *Enterobacterales* infections: Hydrolysable β -lactams increase phenotypic resistance

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1 Background

Exposure to certain β -lactams can induce overexpression of chromosomally-encoded AmpC β -lactamases in ESCPM bacteria (*Enterobacter cloacae*, *Klebsiella aerogenes*, *Serratia marcescens*, *Citrobacter freundii*, *Providencia spp.*, and *Morganella morganii*), decreasing susceptibility [1]. While this mechanism is well understood, clinical impact and extent to which treatment drives resistance remains unclear due to lack of randomized controlled trials and confounding in observational studies [2].

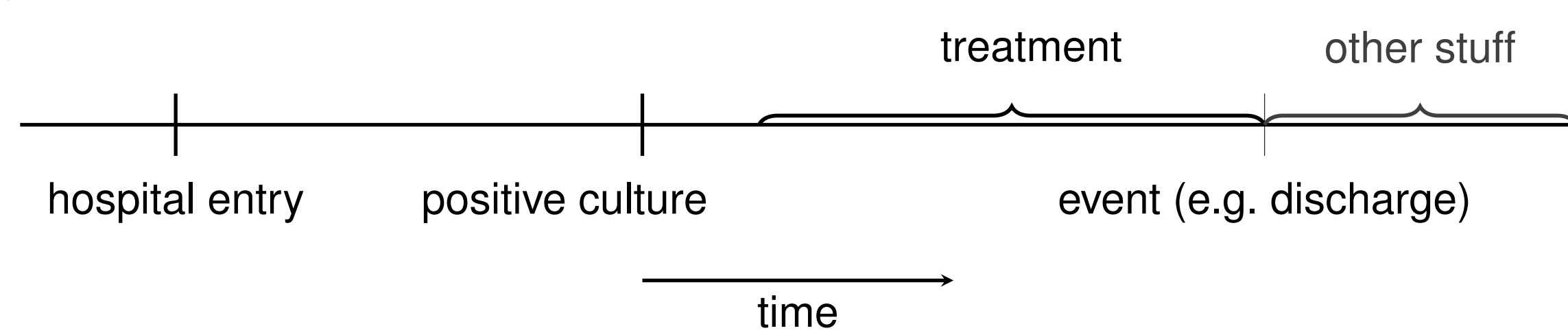
2 Methods

We emulate target trials [3] using electronic health records of adult inpatients with ESCPM infections from a tertiary-care centre, comparing three strategies:

- inadequate antibiotics (by AmpC hydrolysable β -lactams),
- carbapenems/cefepime (current recommendations),
- DNA inhibitors.

Outcomes include length of stay, in-hospital mortality, microbiological relapse, and phenotypic resistance.

a) Single-culture – clinical outcomes



b) Repeated-culture – resistance selection

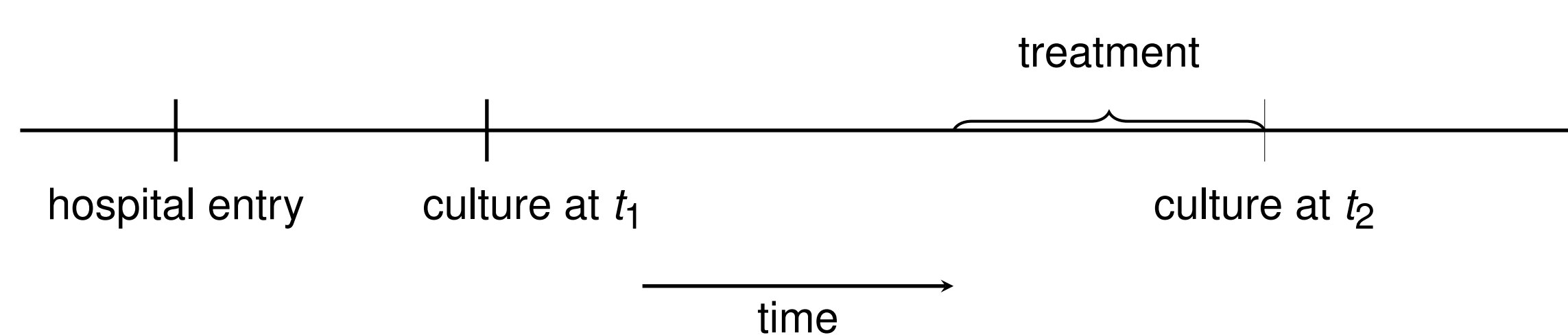


Figure 1: Timelines of the single- and repeated culture target trials.

We adjusted for confounding by patient demographics, comorbidities, infection site, and severity using inverse probability weighting (IPW).

3 Results

Among 3'962 eligible encounters, 915 with repeated cultures were included in the resistance analysis. After IPW, covariate balance was achieved (standardized mean differences ≤ 0.1).

Inadequate therapy was associated with increased odds of new AmpC-like phenotype (adjusted odds ratio (aOR) 2.04, 95%-CI: 1.10 – 3.80) and persistence of resistance (aOR 1.63, 95%-CI: 0.87 – 3.08).

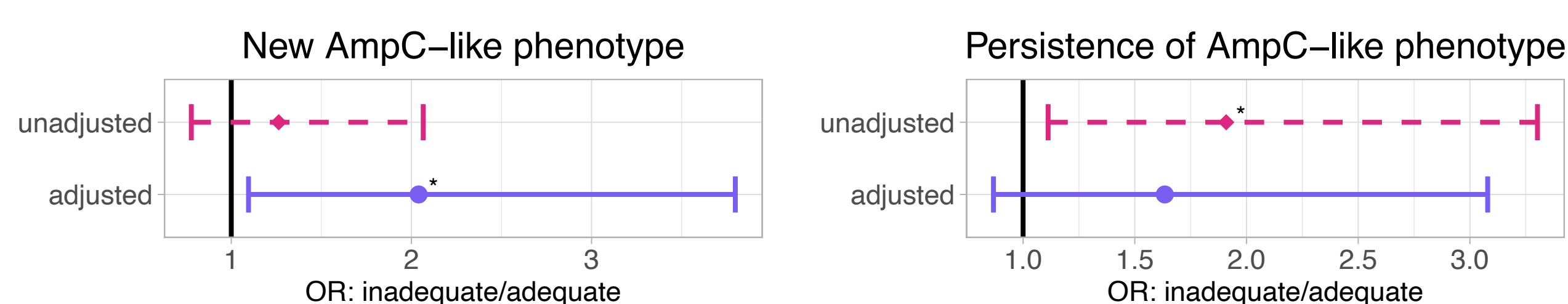


Figure 3: Effect of inadequate treatment on the occurrence of new AmpC-like phenotype and its persistence

3 Results

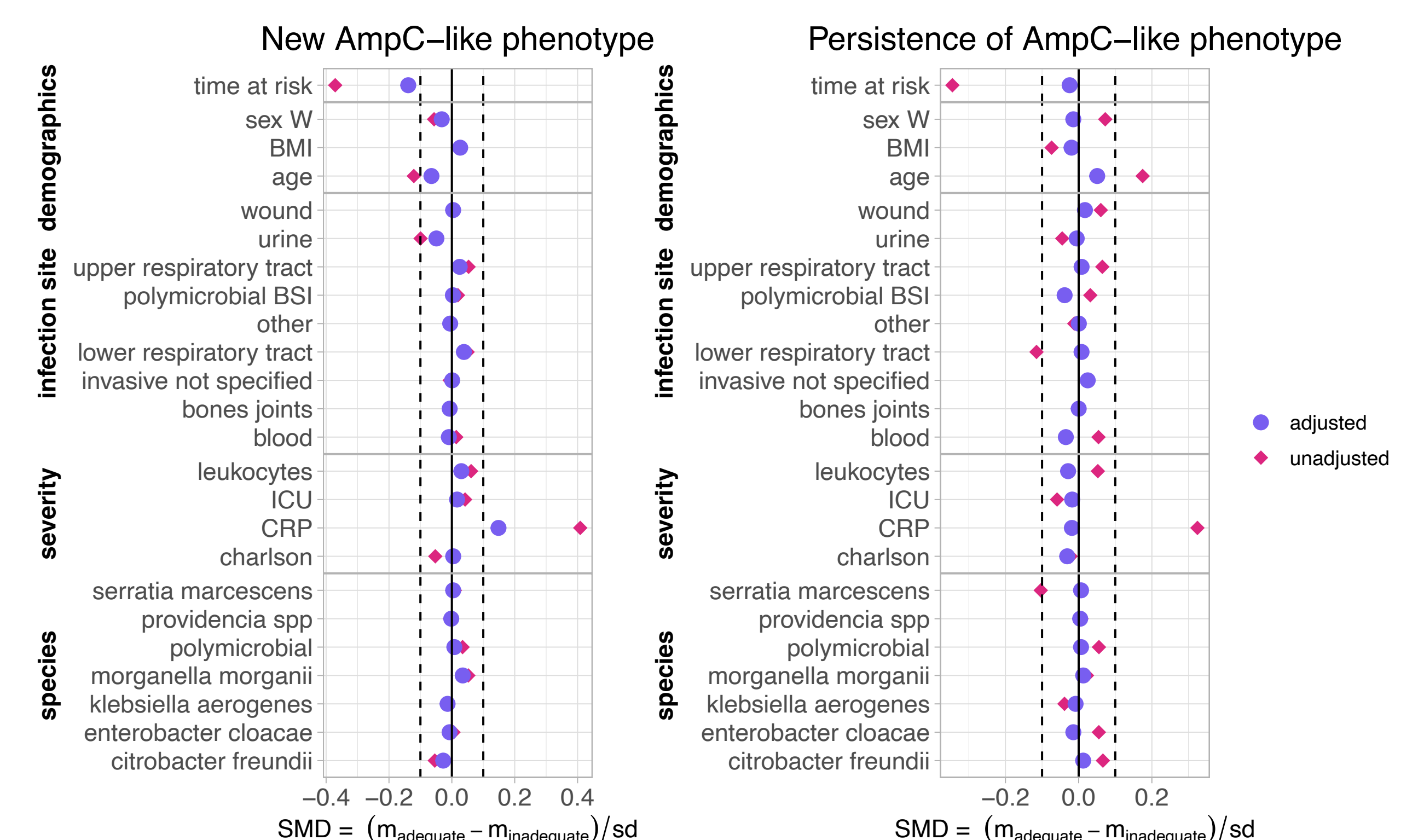


Figure 4: Standardized mean differences between treatment arms before and after adjustment.

In a stratified analysis, meropenem was associated with lower odds of reduced susceptibility compared with cefepime (aOR 0.21, 95%-CI: 0.06 – 0.70).

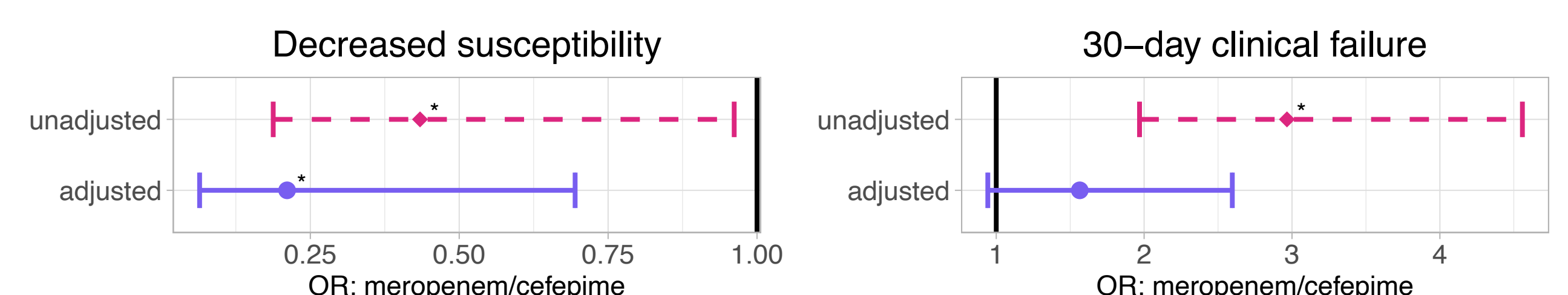


Figure 4: Treatment effect of meropenem compared to cefepime on decreased susceptibility and clinical failure.

Clinical outcomes – mortality, ICU admission, and length of stay – did not differ substantially, and microbiological relapse was not associated with treatment strategy.

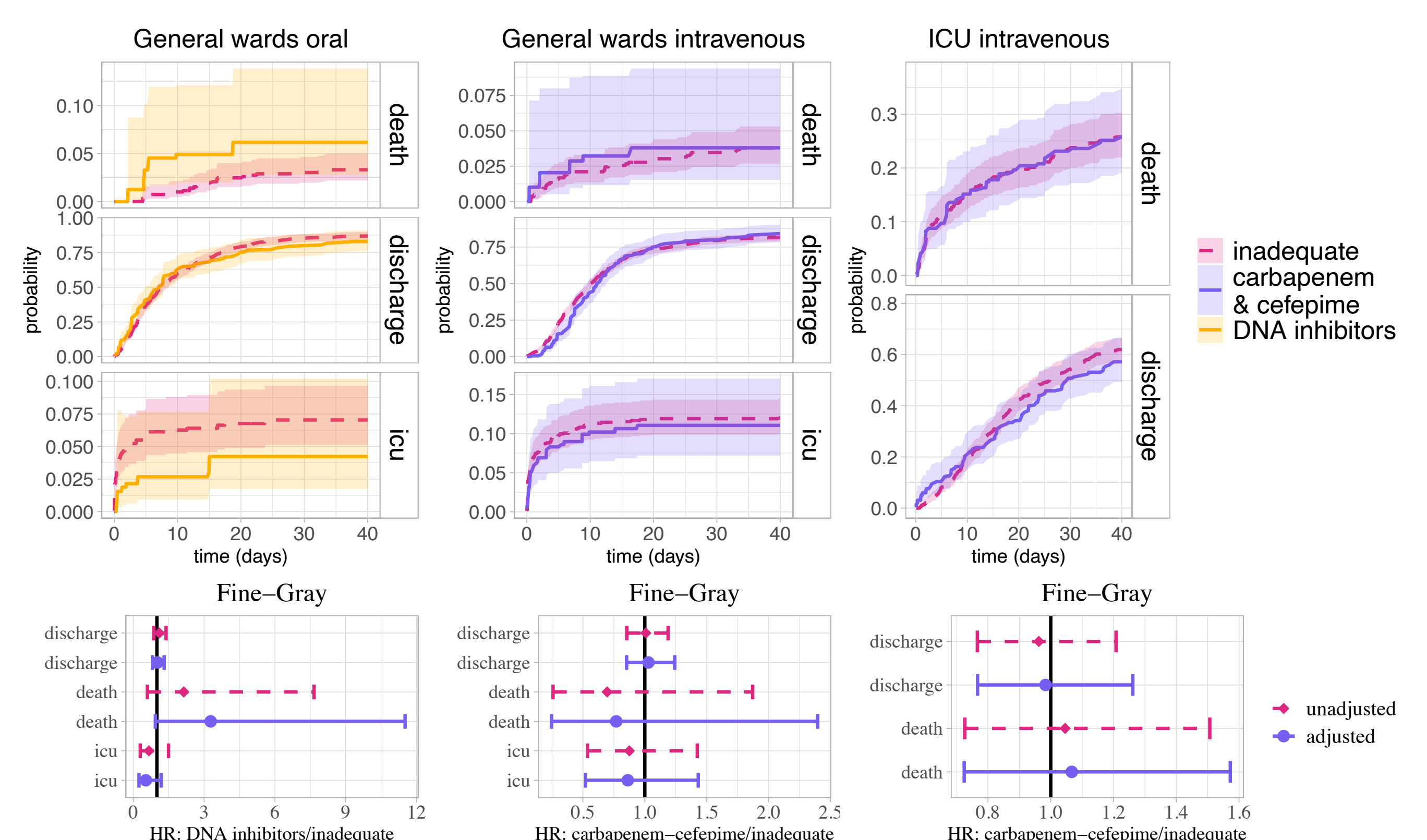


Figure 5: Effects of empirical treatment on discharge, death, and ICU admission rates.

4 Discussion

The findings suggest that while short-term clinical outcomes may be similar, inappropriate use of β -lactams is associated with decreased susceptibility, emphasising the need of antimicrobial stewardship.

References

- [1] Pranita D Tamma, Yohei Doi, et al., “A primer on AmpC β -lactamases: necessary knowledge for an increasingly multidrug-resistant world”, *Clinical infectious diseases* 69.8 (2019), pp. 1446–1455.
- [2] Adam G Stewart, David L Paterson, et al., “Meropenem versus piperacillin-tazobactam for definitive treatment of bloodstream infections caused by AmpC β -lactamase-producing *Enterobacter spp.*, *Citrobacter freundii*, *Morganella morganii*, *Providencia spp.*, or *Serratia marcescens*: a pilot multicenter randomized controlled trial (MERINO-2)”, *Open forum infectious diseases*, vol. 8, 8, Oxford University Press US, 2021, ofab387.
- [3] Miguel A. Hernán and James M. Robins, “Using Big Data to Emulate a Target Trial When a Randomized Trial Is Not Available”, *American Journal of Epidemiology* 183.8 (Mar. 2016), pp. 758–764.